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Letter

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Additional Insights between Fermi−Löwdin Orbital SIC and the Localization Equation Constraints in SIC-DFT

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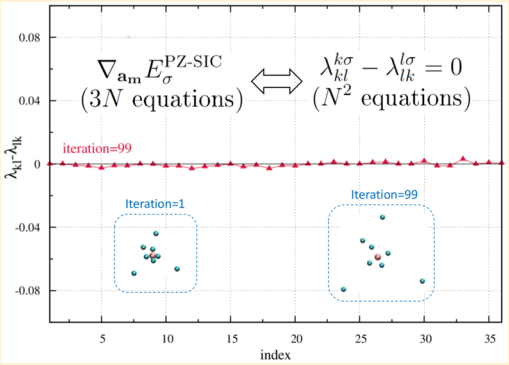
Supporting Information

ABSTRACT: This letter highlights additional mathematical relationships between the Fermi−Löwdin orbital self-interaction correction (FLO-SIC) formalism and the localization equation constraints in SIC-DFT. We demonstrate this relationship *analytically* by highlighting symmetries in the mathematical expression for the gradient of $E^{\mathrm{PZ-SIC}}$, which has not been previously shown in the scientific literature. To complement our analytical derivation, we also present additional numerical tests that allow us to investigate a possible accelerated-convergence technique that could be used when solving the iterative FLO-SIC equations. Taken together, our results highlight the importance of satisfying the localization equation constraints for obtaining accurate DFT energies, which we demonstrate are nearly satisfied in the FLO-SIC formalism.



There has been recent renewed interest in self-interaction correction (SIC) approaches within various applications of Kohn−Sham density functional theory (DFT).[1−8] Originally suggested by Perdew and Zunger (PZ) in 1981,[9] two recent variational modifications of the PZ-SIC approach have emerged in the past four years: (1) a unitary transformation approach for optimizing complex-valued orbitals via an Edminston−Ruedenberg-like localization scheme,[10] as proposed by Jonnson and co-workers,[7−11] and (2) a constrained unitary transformation scheme parametrized by geometric centroids in which real Fermi−Löwdin orbitals are used instead.[12−15] The Fermi−Löwdin orbital self-interaction correction (FLO-SIC) is particularly noteworthy since this approach requires the optimization of $3N$ parameters (instead of solving N^2 localization equations as required in the original PZ-SIC approach), resulting in significant computational savings. Very recently in 2017−2018, Jackson and co-workers[1,16] presented *numerical* tests of the FLO-SIC approach to investigate its connection to the localization equation (LE)[17−20] constraints required in the original SIC approach. Specifically, these prior works found that the original SIC and FLO-SIC approaches agreed for systems containing a small number (N) of electrons since $3N$ is larger than $N(N + 1)/2$ for small values of N. In these numerical tests, Jackson and co-workers noticed that their results were intriguing and commented that "because (FLO-SIC) avoids the LE constraints, it is not obvious that FLO-SIC should lead to the same optimal solutions as methods that explicity satisfy those constraints. Yet for the atoms from Li to Ar, this appears to be so."[1] This letter sheds additional light on this conundrum by providing a new *analytical* derivation that provides a connection between FLO-SIC and the LE constraints. To complement our analytical derivation, we also present additional numerical tests that highlight this mathematical connection between the FLO-SIC and LE approaches. Finally, our analytical and numerical findings allow us to provide a possible accelerated-convergence technique[21] that could be used when solving the iterative FLO-SIC equations.

In short, the FLO-SIC approach introduces a constrained unitary transformation on the Kohn−Sham orbitals that is built upon a procedure similar to a geometry optimization where numerical optimization algorithms are used (e.g., BFGS[22−25] or conjugate gradient[26]) that effectively minimize the forces acting on the atoms. The removal of self-interaction errors enables improved predictions of ground-state properties, atomization energies, ionization potentials, charge transfer properties, and magnetic properties.[27] The FLO-SIC methodology for the removal of self-interaction errors is as follows: A conventional single point energy calculation yields the Kohn−Sham molecular orbitals (KS MOs). In our implementation, we commence with spatially localized orbitals that resemble localized Wannier orbitals using the Foster−Boys algorithm[28] which we only use as an initial guess to compute initial centroids, $\mathbf{a}_m$. The next step is to construct (nonorthogonal) Fermi orbitals followed by (orthogonal) Löwdin orbitals $\phi_{i\sigma}$ that are used to compute orbital densities, $\rho_{i\sigma} = |\phi_{i\sigma}|^2$. The SIC energy, $E^{\mathrm{PZ-SIC}}$, and its corresponding gradient, $\nabla_{\mathbf{a}_m} E_\sigma^{\mathrm{PZ-SIC}}$, is then computed using those orbital densities, Fermi orbital coefficients, Löwdin orbital coefficients, Fermi orbital eigenvalues, and corresponding eigenvectors. A detailed

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explanation of this methodology is provided by Pederson and co-workers.[13] The optimization process consists of using a numerical algorithm (e.g L-BFGS-b[29−31] or gradient algorithms) that minimizes $E^{\mathrm{PZ-SIC}}$, resulting in $\nabla_{\mathbf{a}_m} E_\sigma^{\mathrm{PZ-SIC}} \to \vec{0}$. As a result of this process, the geometric centroids $\mathbf{a}_m$ rearrange themselves to optimized locations around the molecule. We have implemented this method in an in-house version of the open-source quantum chemistry software, NWChem.[32] The benchmarks and performance of our implemented FLO-SIC approach will be published in another paper.

Earlier studies[17−20] have suggested that the orbitals used to construct the SIC energy should satisfy the $O(N^2)$ localization equations (LEs),[17−20] given by

$$\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma} = 0,\ (N^2 \text{ equations}) \tag{1}$$

where $\lambda_{lk}^{k\sigma}$ is

$$\lambda_{lk}^{k\sigma} = \langle \phi_{l\sigma} | V_{k\sigma}^{\mathrm{SIC}} | \phi_{k\sigma} \rangle \tag{2}$$

and the SIC potential, $V_{k\sigma}^{\mathrm{SIC}}$, is defined as

$$V_{k\sigma}^{\mathrm{SIC}} = \frac{\delta E^{\mathrm{PZ-SIC}}}{\delta \rho_{k\sigma}} = -\frac{\delta E_{xc}^{\mathrm{approx}}[\rho_{k\sigma}, 0]}{\delta \rho_{k\sigma}} - \int \mathrm{d}\mathbf{r}\, \frac{\rho_{k\sigma}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \tag{3}$$

Here $\rho_{k\sigma} = |\phi_{k\sigma}|^2$ are orbital electron densities for the kth electron with a given polarization σ. According to Pederson et al.,[18] the LEs are an extended version of the Edmiston and Ruedenberg[10] method for the localization of orbitals, which are obtained from minimizing the total energy including the SIC correction.

In a subsequent series of papers, Pederson et al.[12,13,15,21,33] suggested that one could avoid solving the expensive LEs by minimizing the forces acting on the centroids that arise from $E^{\mathrm{PZ-SIC}}$, leading to the condition $\nabla_{\mathbf{a}_m} E_\sigma^{\mathrm{PZ-SIC}} \to \vec{0}$. Following this new approach, one solves only $3N$ equations instead of N^2 equations, which makes this approach significantly more computationally efficient.

To more clearly show the connection between FLO-SIC and the LE constraints, we briefly review the relevant expressions provided by Pederson et al.[12,15] for the derivative of the total SIC energy with respect to the centroids $\mathbf{a}_m$, $\nabla_{\mathbf{a}_m} E_\sigma^{\mathrm{PZ-SIC}}$:

$$\nabla_{\mathbf{a}_m} E_\sigma^{\mathrm{PZ-SIC}} = \sum_{k=1}^{N} \sum_{\substack{l=1 \\ (l \neq k)}}^{N} \lambda_{kl}^{k\sigma} (\vec{\Delta}_{lk,m}^{1\sigma} + \vec{\Delta}_{lk,m}^{3\sigma}),$$

$$(3N \text{ equations}) \tag{4}$$

where $\vec{\Delta}_{lk,m}^{1\sigma}$ and $\vec{\Delta}_{lk,m}^{3\sigma}$ are vector quantities defined as

$$\vec{\Delta}_{lk,m}^{1\sigma} = \sum_{\alpha\beta n} (\nabla_{\mathbf{a}_m} S_{nm}) T_{\alpha m}^{\sigma} T_{\beta n}^{\sigma} \frac{T_{\alpha k}^{\sigma} T_{\beta l}^{\sigma} - T_{\alpha l}^{\sigma} T_{\beta k}^{\sigma}}{\sqrt{Q_\alpha^\sigma Q_\beta^\sigma}} \tag{5}$$

$$\vec{\Delta}_{lk,m}^{3\sigma} = -\frac{1}{2} \sum_{\alpha\beta n} \nabla_{\mathbf{a}_m} S_{nm} (T_{\beta n}^{\sigma} T_{\alpha m}^{\sigma} + T_{\beta m}^{\sigma} T_{\alpha n}^{\sigma})$$

$$(T_{\alpha k}^{\sigma} T_{\beta l}^{\sigma} - T_{\alpha l}^{\sigma} T_{\beta k}^{\sigma}) \frac{\sqrt{Q_\beta^\sigma} - \sqrt{Q_\alpha^\sigma}}{\sqrt{Q_\alpha^\sigma Q_\beta^\sigma} (\sqrt{Q_\alpha^\sigma} + \sqrt{Q_\beta^\sigma})} \tag{6}$$

S_{nm} is the Fermi-orbital (FO) overlap matrix, and Q_α^σ and $T_{\alpha m}^{\sigma}$ are the eigenvalues and matrix of eigenvectors, respectively, of the FO overlap matrix. Upon close examination of these expressions, we found that $\vec{\Delta}_{lk,m}^{1\sigma}$ and $\vec{\Delta}_{lk,m}^{3\sigma}$ are antisymmetric:

$$\vec{\Delta}_{lk,m}^{1\sigma} = -\vec{\Delta}_{kl,m}^{1\sigma} \tag{7}$$

and

$$\vec{\Delta}_{lk,m}^{3\sigma} = -\vec{\Delta}_{kl,m}^{3\sigma} \tag{8}$$

Using eqs 7 and 8, we can reduce eq 4 to

$$\nabla_{\mathbf{a}_m} E_\sigma^{\mathrm{PZ-SIC}} = \sum_{k>l} (\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma})(\vec{\Delta}_{lk,m}^{1\sigma} + \vec{\Delta}_{lk,m}^{3\sigma}),$$

$$(3N \text{ equations}) \tag{9}$$

Equation 9 is a vector equation with $3N$ components (which correspond to N independent centroids, each of which resides in 3 dimensions). Figure 1 shows a diagrammatic representa-

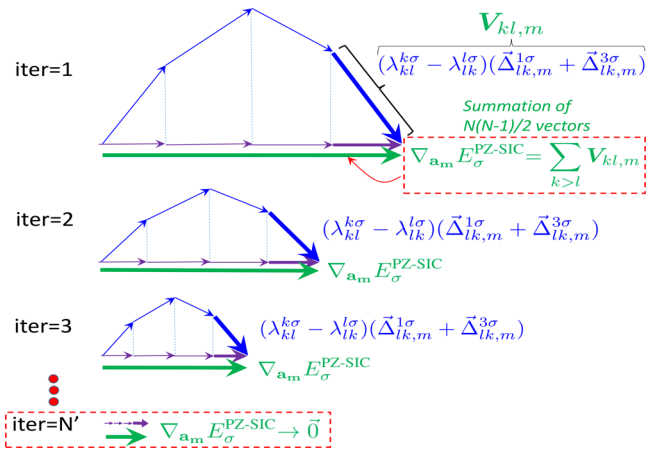

Figure 1. Schematic showing the magnitude of the vector, $\nabla_{\mathbf{a}_m} E_\sigma^{\mathrm{PZ-SIC}}$, at each iteration step. $\nabla_{\mathbf{a}_m} E_\sigma^{\mathrm{PZ-SIC}}$ is composed of the vectorial summation: $\sum_{k>l} (\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma})(\vec{\Delta}_{lk,m}^{1\sigma} + \vec{\Delta}_{lk,m}^{3\sigma})$ (cf. eq 9). The contribution from each individual vector $(\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma})(\vec{\Delta}_{lk,m}^{1\sigma} + \vec{\Delta}_{lk,m}^{3\sigma})$ along the direction of $\nabla_{\mathbf{a}_m} E_\sigma^{\mathrm{PZ-SIC}}$ is indicated by the purple vectors. The magnitude of the projected vectors along the gradient (e.g., the purple vectors) is proportional to $\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma}$ and decreases during the iteration process. At the end of the iteration/optimization we obtain $|\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma}| < \delta$, where δ is a numerically specified threshold (typically ≈0.015 hartree in this work).

tion of eq 9 throughout this numerical minimization procedure. In this schematic, the contribution from each individual vector $(\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma})(\vec{\Delta}_{lk,m}^{1\sigma} + \vec{\Delta}_{lk,m}^{3\sigma})$ along the direction of $\nabla_{\mathbf{a}_m} E_\sigma^{\mathrm{PZ-SIC}}$ is indicated by the purple vectors in Figure 1. The magnitude of the projected vectors along the gradient (e.g., the purple vectors) is proportional to $\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma}$ and decreases during the iteration process. As the iteration/minimization proceeds, the magnitude of $\nabla_{\mathbf{a}_m} E_\sigma^{\mathrm{PZ-SIC}}$ (and $|\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma}|$) decreases toward zero, yielding the following relationship:

$$\text{If } \nabla_{\mathbf{a}_m} E_\sigma^{\mathrm{PZ-SIC}} \to \vec{0}, \text{ then:} |\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma}| < \delta \tag{10}$$

where δ is a numerically specified threshold (typically ≈0.015 hartree in this work). This is the most salient result of our paper, and the expressions for $\lambda_{kl}^{k\sigma}$ in eqs 9 and 10 resemble the localization equations in eq 1. This mathematical connection has not been previously shown or demonstrated in the

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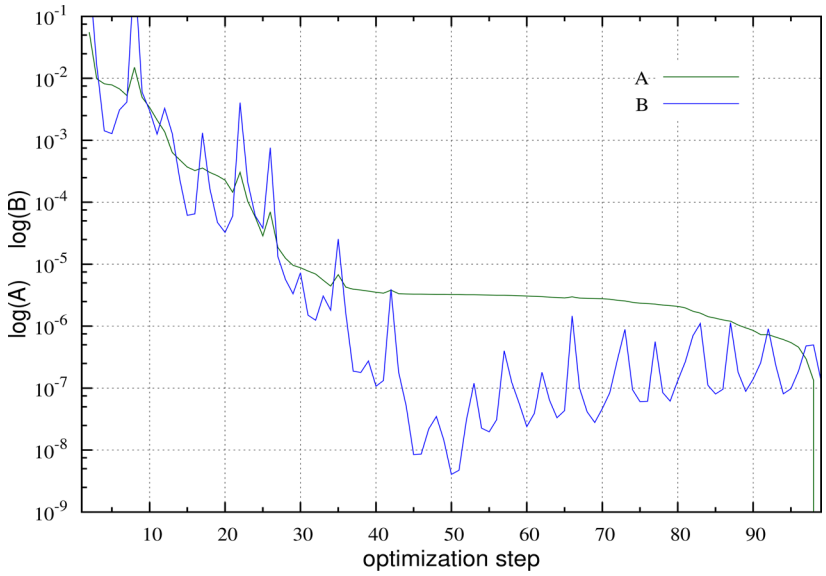

Figure 2. Optimization process for Ar (Z = 18) showing: (1) the SIC energy convergence (green curve) and (2) the SIC energy gradient decreasing to a negligible value (blue curve). The labels A and B denote the expressions: A = $E^{\mathrm{PZ-SIC}} - E^{\mathrm{PZ-SIC}}_{\mathrm{final}}$, B = $\sum_m \nabla_{\mathbf{a}_m} E^{\mathrm{PZ-SIC}} \cdot \nabla_{\mathbf{a}_m} E^{\mathrm{PZ-SIC}}$. Calculations are carried out with the LDA-PW91[37]/FLO-SIC exchange-correlation approach and the def2-TZVP[38] basis set.

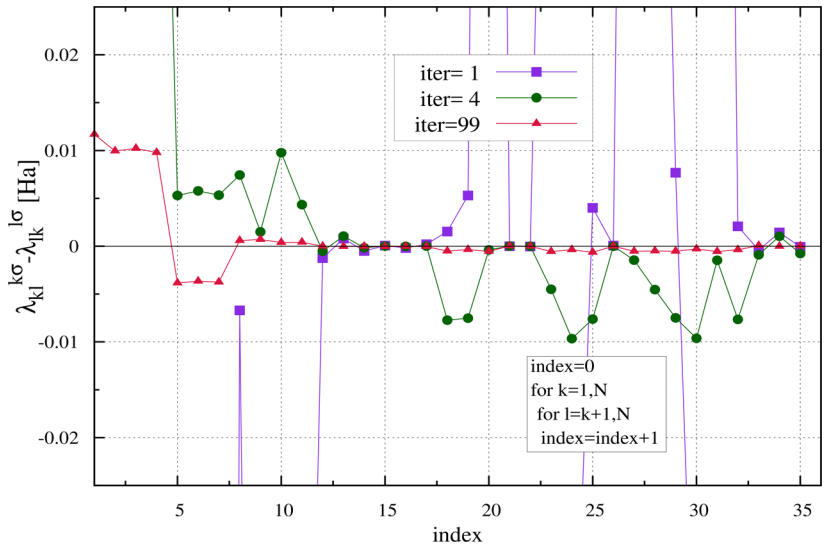

Figure 3. Evolution of $\lambda_{kl} - \lambda_{lk}$ at three stages of the FLO-SIC optimization process: (1) beginning (iteration = 1), (2) intermediate (iteration = 4), and (3) end (iteration = 99). Calculations are carried out on argon (number of electrons = 18) with the LDA-PW91[37]/FLO-SIC exchange-correlation approach and the def2-TZVP[38] basis set. The horizontal axis denotes a linear indexing scheme for plotting all elements of the matrix λ_{lk}, where index = 1,2, ..., $N(N-1)/2$ (where N = 9, the number of centroids), which is obtained according to the pseudocode shown in the inset of the graph. The final optimized values (e.g., iter = 99) give an RMS deviation of 3.7×10^{-3} hartree.

literature, and eq 9 presents this connection *analytically*, whereas previous approaches have only shown this with numerical tests. We emphasize that our derivation certainly does not constitute a "proof" of the equivalence between FLO-SIC and the original LEs in SIC-DFT; rather, our intention is to provide additional insight (i.e., eqs 9 and 10) into why the numerical FLO-SIC procedure appears to work so well in certain select cases, as previously noted by other researchers. We now give additional numerical tests of eq 10 by applying the FLO-SIC formalism on two systems: (1) an argon atom and (2) the benzene molecule.

Figure 2 shows the results of the $E^{\mathrm{PZ-SIC}}$ optimization process performed on Argon (number of electrons = 18, number of centroids = 9). We observe that the SIC energy, $E^{\mathrm{PZ-SIC}}$, monotonically converges with the energy gradients decreasing to a negligible value, $\nabla_{\mathbf{a}_m} E^{\mathrm{PZ-SIC}}_{\sigma} \to \vec{0}$. Figure 3 plots all of the unique $\lambda_{kl} - \lambda_{lk}$ combinations at different optimization steps, which show that the LEs are nearly satisfied (i.e., $\lambda_{kl} - \lambda_{lk} \approx 0$) at the end of the optimization procedure, giving a root-mean-square (RMS) deviation of 3.7×10^{-3} hartree, which is comparable to the previous results obtained by Kao et al.[1] In Figure S1 of the Supporting Information, we compare several different minimization algorithms (e.g., L-BFGS-b,[29,34,35] conjugate gradient,[36] and Powell[26]) as well as different initial conditions to show that our implementation of the FLO-SIC methodology is correct and yields RMS values that are similar to the values obtained by Kao et al.[1] In the Supporting Information, we also provide

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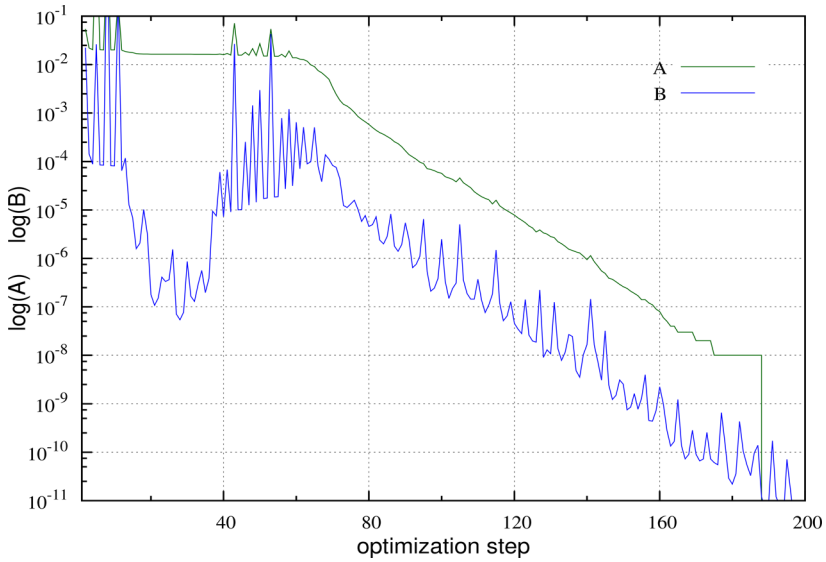

Figure 4. Optimization process for benzene (number of electrons = 42) showing: (1) the SIC energy convergence (green curve) and (2) the SIC energy gradient decreasing to a negligible value (blue curve). The labels A and B denote the expressions: A = $E^{\mathrm{PZ-SIC}} - E^{\mathrm{PZ-SIC}}_{\mathrm{final}}$, B = $\sum_m \nabla_{\mathbf{a}_m} E^{\mathrm{PZ-SIC}} \cdot \nabla_{\mathbf{a}_m} E^{\mathrm{PZ-SIC}}$. Calculations are carried out with the LDA-PW91[37]/FLO-SIC exchange-correlation approach and the def2-TZVP[38] basis set.

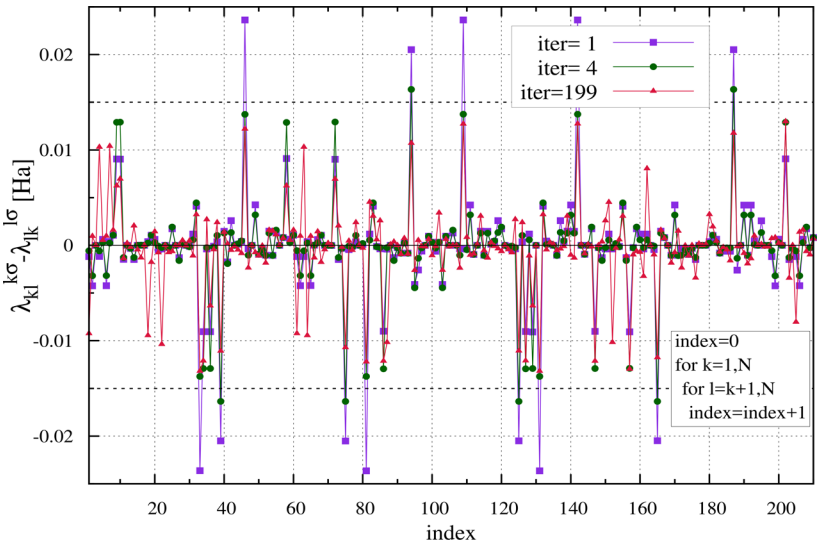

Figure 5. Evolution of $\lambda_{kl} - \lambda_{lk}$ at three stages of the FLO-SIC optimization process: (1) beginning (iteration = 1), (2) intermediate (iteration = 4), and (3) end (iteration = 199). Calculations are carried out on benzene with the LDA-PW91[37]/FLO-SIC exchange-correlation approach and the def2-TZVP[38] basis set. The horizontal axis denotes a linear indexing scheme for plotting all elements of the matrix λ_{lk}, where index = 1,2, ..., $N(N-1)/2$ (where N = 21, the number of centroids), which is obtained according to the pseudocode shown in the inset of the graph. The final optimized values (e.g., iter = 199) give an RMS deviation of 4.6×10^{-3} hartree.

detailed plots of the x, y, and z components of the vectors $\vec{\Delta}^{1\sigma}_{lk,m}$ and $\vec{\Delta}^{3\sigma}_{lk,m}$ for all centroids m = 1,2, ..., N. The plots of the vector $\vec{\Delta}^{1\sigma}_{lk,m}$ consistently show large peaks appearing at all iterations for all centroids m and directions x, y, and z. On the other hand, the plots of the vector $\vec{\Delta}^{3\sigma}_{lk,m}$ show convergence to negligible values at the end of the optimization process (iteration number = 99). In the Supporting Information, we have included plots of the projections of the individual vectors, $\hat{u} \cdot \mathbf{V}_{kl,m}$, from the summation $\nabla_{\mathbf{a}_m} E^{\mathrm{PZ-SIC}}_{\sigma} = \sum_{k>l} \mathbf{V}_{kl,m}$ along the direction of the total gradient $\hat{u} = \nabla_{\mathbf{a}_m} E^{\mathrm{PZ-SIC}}_{\sigma} / |\nabla_{\mathbf{a}_m} E^{\mathrm{PZ-SIC}}_{\sigma}|$ during the first four stages of the optimization process. These plots collectively show that indeed the magnitude of each of the projected vectors (i.e., the purple vectors in Figure 1), $|\hat{u} \cdot \mathbf{V}_{kl,m}|$, decreases consistently as the optimization process progresses. Additionally, the plots in Figures S9−S11 show that large peaks appear for indices that correspond to the centroids closest to the Argon atom (these are composed of four centroids that form a tetrahedron around the Argon atom and a centroid that is nearly centered at the Argon atom). As such, one can conclude that the reason the gradients of $E^{\mathrm{PZ-SIC}}$ become negligible at the end of optimization, $\nabla_{\mathbf{a}_m} E^{\mathrm{PZ-SIC}}_{\sigma} \to \vec{0}$, is primarily due to the $\lambda^{k\sigma}_{kl} - \lambda^{l\sigma}_{lk}$ coefficient that multiplies the quantity $\vec{\Delta}^{1\sigma}_{lk,m} + \vec{\Delta}^{3\sigma}_{lk,m}$ within the summation in eq 9. Similar results are shown in Figures 4 and 5 for the case of benzene. Since benzene contains a larger number of centroids (e.g., N = 21), the energy optimization process is more complex and requires a larger number of optimization steps. Also, in Figure 5, the curves for $\lambda^{k\sigma}_{kl} - \lambda^{l\sigma}_{lk}$ contain a larger number of points

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Table 1. Minimization of E^{PZ-SIC} Using Two Objective Functions: (1) E^{PZ-SIC} and (2) R (eq 11) for the Argon Atom.[a]

objective function: E^{PZ-SIC}			objective function: R		
iteration	E^{PZ-SIC} (hartree)	J_E (hartree/Å)^2	iteration	E^{PZ-SIC} (hartree)	J_R (1/Å^2)
1	−0.83718916	37.94886604	1	−0.83718916	1.57341718
2	−2.47894653	0.54751964	2	−2.58768291	0.00083182
3	−2.57047112	0.01640812	3	−2.58626187	0.00065899
4	−2.57386152	0.00141868	4	−2.53330545	0.00304196
⋮	⋮	⋮	⋮	⋮	⋮
22	−2.59029556	0.00044420	22	−2.36497311	0.00127391
23	−2.59040528	0.00005109	23	−2.39098002	0.00015929
24	−2.59044427	0.00001263	24	−2.11037838	0.20891197
25	−2.59045786	0.00001463	25	−2.37841108	0.00043249

[a]All calculations were carried out with LDA-PW91/FLO-SIC exchange-correlation approach and the def2-TZVP basis set.

that show a complicated structure. We observe that larger peaks decrease in value as the iteration step increases and, therefore, $|\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma}| < 0.015$ Ha. This is still a negligible value that approximately satisfies the constraint imposed by the LEs while also simultaneously causing $\nabla_{\mathbf{a}_m} E_\sigma^{PZ-SIC}$ to decrease to small values at the end of the optimization process. In Figure S2 of the Supporting Information, we plot all of the unique $\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma}$ combinations for a larger pentacene molecule with 72 centroids (benzene contains only 21 centroids), where we obtain $|\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma}| < 0.015$ hartree and an RMS deviation of 2.6 $\times 10^{-3}$ hartree, which is similar in accuracy to the results for benzene. We also note that while these RMS values are not exactly zero, the self-interaction-corrected energies (as well as ionization potentials and MO energy levels) that result from these optimizations are still extremely accurate and are comparable to standard PZ-SIC approaches.

While eq 10 and its connection to the LE constraint in SIC is the most important point of this paper, we also examined other proposed methods for minimizing E^{PZ-SIC}. Very recently in 2016, a previous study suggested a less expensive approach for minimizing the FLO-SIC energy[21] and stated: "For physical reasons, we assume that each of the Fermi orbitals with the correct Fermi orbital descriptors contains exactly one electron, i.e. the corresponding Löwdin eigenvalues are all equal to one. Hence, the optimization of the Fermi orbital descriptors via the minimization of the SIC energy could be replaced by a minimization of the difference from the Löwdin eigenvalues to one..." This prior suggestion is worth investigating since it bypasses the computation of the Coulomb and exchange-correlation integrals in λ_{lk} (cf. eq 2) and is, therefore, extremely computationally efficient. To test this suggestion, we can construct a new objective function, R:

$$R = \sum_{\alpha=1}^{N} \omega_\alpha (Q_\alpha - 1)^2 \tag{11}$$

Minimization of this objective function requires its gradient to go to zero:

$$\nabla_{\mathbf{a}_m} R = 2 \sum_{\alpha=1}^{N} \omega_\alpha (Q_\alpha - 1) \nabla_{\mathbf{a}_m} Q_\alpha \to \vec{0} \tag{12}$$

An expression for $\nabla_{\mathbf{a}_m} Q_\alpha$ is provided in ref 21, and ω_α is a general weighting coefficient that we take, for simplicity, to be equal to $1/N$. The objective function in eq 11 is a standard approach for reducing a multiobjective optimization problem into a single objective optimization problem[39−41] to test the suggestions in ref 21. To clearly depict the minimization process of these two approaches, we calculate the scalar quantities, J_E and J_R in Table 1, given by

$$J_E = \sum_{m=1}^{N} \nabla_{\mathbf{a}_m} E^{PZ-SIC} \cdot \nabla_{\mathbf{a}_m} E^{PZ-SIC} \quad \text{(for objective function } E^{PZ-SIC}) \tag{13}$$

$$J_R = \sum_{m=1}^{N} \nabla_{\mathbf{a}_m} R \cdot \nabla_{\mathbf{a}_m} R \quad \text{(for objective function } R) \tag{14}$$

together with their corresponding SIC energy values for the argon atom.

To provide a fair comparison for each of these two approaches, we used the same initial centroids, $\mathbf{a}_{m,0}$, and molecular orbitals as initial values for the iteration process. While the results of Table 1 clearly show that the optimization procedures for both E^{PZ-SIC} and R properly converge (i.e., both J_E and J_R approach zero), a very incorrect SIC energy of −2.378 hartree is obtained when only R is used (the correct SIC energy is −2.590 hartree). While we have shown this for the Argon atom as a specific example, we have obtained similar convergence behavior for both J_E and J_R in other chemical systems, which emphasizes the importance of satisfying the LE constraints via eq 9. Nonetheless, since the R objective function is still significantly less expensive than directly minimizing E^{PZ-SIC}, one can still utilize this approach as an initial guess to partially accelerate the convergence of the FLO-SIC procedure. In Figure 6, we compare the convergence of the conventional E^{PZ-SIC} minimization procedure (denoted as "Opt E") against a hybrid approach (denoted as "Opt Q + Opt E") where the first five iterations minimize R and the remaining iterations minimize E^{PZ-SIC}. As can be seen in Figure 6, the Opt Q + Opt E hybrid approach converges at a faster rate than the conventional Opt E approach, yet both of these procedures still converge to the correct SIC energy. More importantly, since the initial steps of the Opt Q + Opt E hybrid approach completely bypasses the calculation of computationally-expensive Coulomb and exchange-correlation integrals (λ_{lk}), this combined approach could be useful for accelerating SIC calculations of larger systems.

In conclusion, this letter gives additional mathematical insight between the FLO-SIC formalism and the localization equation constraints in SIC-DFT. We demonstrate this relationship *analytically* by highlighting symmetries in the mathematical expression for the gradient of E^{PZ-SIC}, and our derivation has not been previously shown or demonstrated in the literature. We emphasize that our derivation does not

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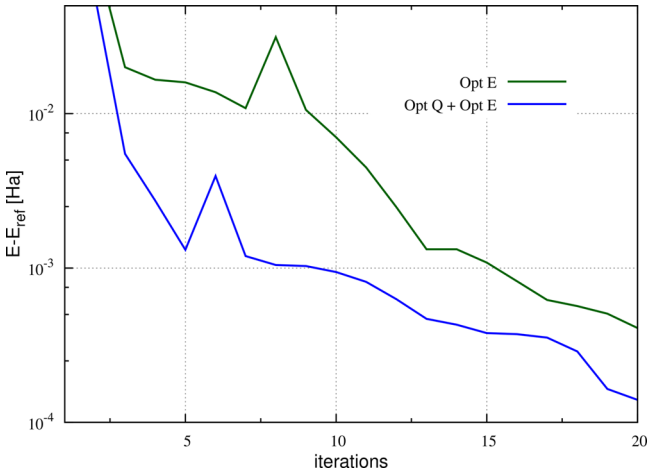

Figure 6. Convergence behavior (measured by calculating $E - E_{\mathrm{ref}}$, where E_{ref} is the ground-state reference SIC energy) for the Argon atom obtained from two different approaches: (1) the conventional $E^{\mathrm{PZ-SIC}}$ minimization procedure (denoted as "Opt E"), and (2) a hybrid approach (denoted as "Opt Q + Opt E") where the first five iterations minimize R and the remaining interations minimize $E^{\mathrm{PZ-SIC}}$. All calculations were carried out with the LDA-PW91/FLO-SIC exchange-correlation approach and the def2-TZVP basis set.

constitute a "proof" of the equivalence between FLO-SIC and the original LEs in SIC-DFT; rather, the derivations in this work (i.e., eqs 9 and 10) give additional mathematical insight into why the numerical FLO-SIC procedure seems to work so well in certain select cases, as previously noted by other researchers. To show additional applications of this connection, we present calculations on the argon atom, benzene molecule, and other chemical systems to show that the LEs are approximately satisfied when $E^{\mathrm{PZ-SIC}}$ is minimized via the FLO-SIC approach. Finally, we also examined a recent suggestion for minimizing $E^{\mathrm{PZ-SIC}}$ via the optimization of Fermi-orbital eigenvalues, Q_α. While this new approach bypasses the computationally expensive Coulomb and exchange-correlation integrals required in the original FLO-SIC approach, we find that this procedure leads to unphysical values of $E^{\mathrm{PZ-SIC}}$ since the LE constraints are not necessarily obeyed. Instead, a hybrid approach that combines this method with the conventional $E^{\mathrm{PZ-SIC}}$ minimization could be useful for accelerating SIC calculations of other systems. Taken together, our results highlight the importance of satisfying the LE constraints (even if the constraints are only approximately satisfied) for obtaining accurate DFT energies, which we demonstrate are approximately satisfied in the FLO-SIC formalism.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpclett.8b02786.

$\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma}$ plots using different minimization algorithms and initial conditions for the Argon atom; $\lambda_{kl}^{k\sigma} - \lambda_{lk}^{l\sigma}$ plots for pentacene; plots of the x, y, and z components of the vectors $\vec{\Delta}_{lk,m}^{1\sigma}$ and $\vec{\Delta}_{lk,m}^{3\sigma}$ for all centroids of the argon atom; projections of individual vectors, $\hat{u} \cdot \mathbf{V}_{kl,m}$, from the summation $\nabla_{\mathrm{a_m}} E_\sigma^{\mathrm{PZ-SIC}} = \sum_{k>l} \mathbf{V}_{kl,m}$ for all centroids of the argon atom (PDF)

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Notes

The authors declare no competing financial interest.

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